

Manas Dhawan

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EDUCATION

University of North Carolina, Chapel Hill (Incoming)

PhD in Statistics and Operations Research

Research Interests: Stochastic Processes, Time Series, and Machine Learning

Chapel Hill, NC

Expected May 2031

University of California, Los Angeles (UCLA)

Bachelor of Science

Major: Mathematics/Economics

GPA: 3.88/4.0

Los Angeles, CA

September 2022 – March 2026

Relevant Coursework: Graduate Probability Theory, Mathematical Statistics, Advanced Econometrics, Time Series, Algorithms and Complexity, Stochastic Processes, R Programming, Mathematical Finance, Machine Learning, Linear Algebra

WORK EXPERIENCE

Blue Bell Private Wealth Management (15 employee Investment Management firm)

Philadelphia, PA

Data Analyst Intern

June 2024 – September 2024

- Leveraged statistical analysis and advanced coding in R to optimize search engine and Google Ad performance, enhancing online visibility and engagement. Formulated comprehensive reports utilizing statistical analysis to optimize SEO.
- Created and implemented mathematical models to solve investment-related and strategic decision-making problems.
- Collaborated closely with team members to research, evaluate, and strategize investment opportunities, focusing on ETFs, options, and closed-end funds to achieve optimal client returns.

Blue Bell Private Wealth Management

Philadelphia, PA

Finance Intern

June 2023 – September 2023

- Led comprehensive research into financial derivatives, evaluating their potential to enhance client services and support.
- Designed and coded algorithms aimed at optimizing retirement portfolios, improving performance and risk management.
- Authored detailed case studies for the company's website, demonstrating successful investment strategies and client achievements, enhancing the firm's online presence, search engine optimization, and credibility.

PROJECTS

Mortgage Rate Forecasting and Volatility Modeling | R, Time Series, GARCH, Monte Carlo

- Developed a full time-series forecasting report on U.S. 30-year mortgage rates, comparing ARIMA, ARDL, Holt-Winters, stochastic simulation, and GARCH volatility models to evaluate both forecast behavior and uncertainty.
- Conducted stationarity testing, STL decomposition, ACF/PACF analysis, Granger-causality-style lag analysis, and residual diagnostics to assess model assumptions and justify modeling choices.
- Simulated future mortgage-rate paths using a stochastic differential equation framework with estimated drift and volatility parameters, then applied ARCH/GARCH modeling to evaluate volatility clustering and persistence in mortgage-rate changes.

ADDITIONAL

Programming Languages: C++(Advanced), Python (Advanced) , R (Advanced) , MATLAB (Intermediate)

Technical Skills: Time Series Analysis, Econometrics, Machine Learning